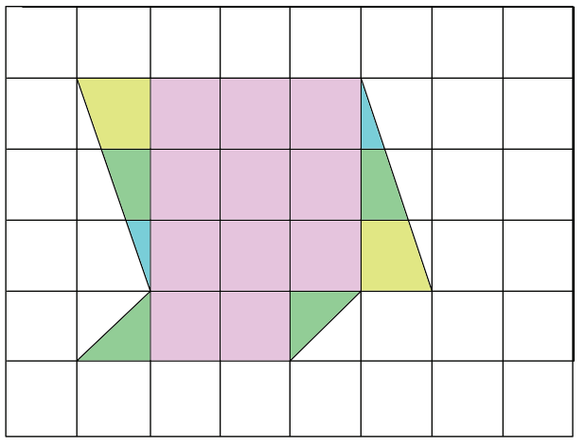
Measure the given area using square method: scale is 1 centimetre = 1 kilometre (area of one small square = 1 square Km)

Solution:



Step 1: Count the number of full squares first = 11

Step 2: Next, count the numbers of the fractional squares

¾ square = 2,

½ square = 4 and

¼ = 2

Step 3: Calculate the total number of squares

11 × 1 = 11

Number of 3/4 square:

2 × 3/4 = 1.5

Number of 1/2 squares:

4 × 1/2= 2 and

number of 1/4 squares:

2 × 1/4 = 0.5)

Step 4: Add all these values

(**11** + **1.5** + **2** + **0.5** = **15** squares)

Step 5: Multiply 15 with the scale of the map

15 × 1 =15 km^2

Hence the area of the given place is = 15 square Km.

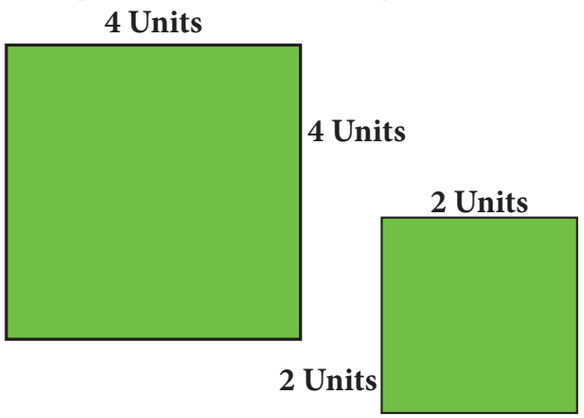
Activity

https://www.makeuseof.com › Internet

Measure the area of any selected village/plot /area of your choice using this link.

9.5 Enlargement and reduction of maps

In the process of compiling maps cartographers are often required to either reduce or enlarge maps. Reduction or enlargement involves change in the size.



An enlargement provides the same map but proportionally larger than the original.

A reduction gives the same map that is proportionally smaller than the original.

The above image or map has been reduced by ½ .The amount that an original image has been enlarged or reduced is called a **scale factor,** or an **enlargement** or **reduction factor**. It is the constant factor by which all dimensions of an object are enlarged or reduced in a map. If shapes have been reduced by half, the scale factor is ½.

The ratio between the area of a map on one scale and its area to another scale is

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